

COUPLED HAMILTON-JACOBI-BELLMAN SYSTEMS

ABSTRACT.

1. INTRODUCTION

2. PROBLEM SETTING

Let $\Omega \subset \mathbb{R}^d$, with $d \geq 2$, be an open, and bounded domain with $C^{1,1}$ boundary. Let σ_1 and σ_2 be constants such that

$$0 < \sigma_1 < \sigma_2.$$

We define the Sobolev spaces

$$W_q := W_q^2(\Omega) \cap \dot{W}_q^1(\Omega), \quad \text{for any } 1 < q < \infty,$$

and

$$W_p := W_p^2(\Omega) \cap \dot{W}_p^1(\Omega),$$

where throughout this paper the exponent p is restricted to satisfy

$$(2.1) \quad p \geq 2 \quad \text{and} \quad \frac{d}{2} < p < \infty.$$

The space $\dot{W}_q^1(\Omega)$ is defined as the closure of $C_c^\infty(\Omega)$ in $W_q^1(\Omega)$, where $C_c^\infty(\Omega)$ denotes the space of infinitely differentiable functions with compact support in Ω .

We denote by $\omega[v]$ the modulus of continuity of a function v , as defined in Definition A.1.

We consider the fully coupled system of HJB equations, given by

$$(2.2) \quad \begin{cases} \sum_{i=1}^d a_i[u_i] \partial_{x_i x_i} u_j - c_j u_j = f_j, & \text{in } \Omega, \\ u_j = 0, & \text{on } \partial\Omega, \end{cases} \quad j = 1, \dots, d,$$

for any $\mathbf{f} = (f_1, \dots, f_d) \in (L_p(\Omega))^d$. Here, for any $v \in W_p^2(\Omega)$ and $i = 1, \dots, d$, the nonlinear coefficients are defined by

$$a_i[v](\mathbf{x}) \in \left\{ \arg \max_{\alpha \in [\sigma_1, \sigma_2]} [\alpha \partial_{x_i x_i} v(\mathbf{x})] \right\} = \begin{cases} \{\sigma_1\}, & \partial_{x_i x_i} v(\mathbf{x}) < 0, \\ [\sigma_1, \sigma_2] & \partial_{x_i x_i} v(\mathbf{x}) = 0, \\ \{\sigma_2\}, & \partial_{x_i x_i} v(\mathbf{x}) > 0, \end{cases} \quad \text{for a.e. } \mathbf{x} \in \Omega.$$

The nonlinearity and the coupling in the system arise entirely through these coefficients. In particular, their dependence on the second derivatives of the unknown functions renders the system both *fully nonlinear* and *strongly coupled*. Moreover, the optimal choice is attained at the boundary of the interval $[\sigma_1, \sigma_2]$, depending on the sign of the curvature.

Fixed point theory is a standard tool for establishing the existence of solutions to coupled systems of PDEs. However, in system (2.2), the coupling occurs through discontinuous and noncompact operators. This prevents a direct application of commonly used fixed point methods based on continuity and compactness, since the required assumptions are not satisfied. To overcome this difficulty, we first introduce and analyze a regularized version of the system, for which fixed point arguments can be applied to prove the existence of solutions.

3. REGULARIZED COUPLED HJB SYSTEM

We now consider a regularized version of the system (2.2), given by

$$(3.1) \quad \begin{cases} \sum_{i=1}^d a_{i,\varepsilon}[u_{i,\varepsilon}] \partial_{x_i x_i} u_{j,\varepsilon} - c_j u_{j,\varepsilon} = f_j, & \text{in } \Omega, \\ u_{j,\varepsilon} = 0, & \text{on } \partial\Omega, \end{cases} \quad j = 1, \dots, d.$$

For $i = 1, \dots, d$, the regularized coefficient operators $a_{i,\varepsilon}: W_q \rightarrow C(\bar{\Omega})$ satisfy the following properties:

1. The operators $a_{i,\varepsilon}$ are continuous and compact.
2. For any $v \in W_q$,

$$(3.2) \quad \lim_{\varepsilon \rightarrow 0} \text{dist} \left(a_{i,\varepsilon}[v](\mathbf{x}), \left\{ \arg \max_{\alpha \in [\sigma_1, \sigma_2]} [\alpha \partial_{x_i x_i} v(\mathbf{x})] \right\} \right) = 0 \quad \text{for a.e. } \mathbf{x} \in \Omega.$$

3. For any $v \in W_q$,

$$(3.3) \quad \sigma_1 \leq a_{i,\varepsilon}[v](\mathbf{x}) \leq \sigma_2 \quad \text{for a.e. } \mathbf{x} \in \Omega.$$

4. The family $\{a_{i,\varepsilon}[v] : v \in W_q\}$ has a modulus of continuity independent of v (in particular, independent of $\|v\|_{W_q^2(\Omega)}$), possibly depending on ε .

The following example provides a standard regularization satisfying the above properties.

Example 3.1 (Regularized coefficients). For any $v \in W_q$, let $\tilde{v}$ denote an extension of v to $\mathbb{R}^d$ satisfying

$$(3.4) \quad \|\tilde{v}\|_{W_q^2(\mathbb{R}^d)} \leq C_{3.4} \|v\|_{W_q^2(\Omega)}.$$

For $i = 1, \dots, d$ and any $v \in W_q$, we define the scaled mollification operator

$$(3.5) \quad \mathcal{S}_{i,\varepsilon}[v](\mathbf{x}) := \frac{1}{\|v\|_{W_q^2(\Omega)} + C_{3.5}} (\eta_\varepsilon * \partial_{x_i x_i} \tilde{v})|_\Omega(\mathbf{x}),$$

where $C_{3.5} > 0$ is a fixed constant and η_ε denotes the standard mollifier. The scaling factor $(\|v\|_{W_q^2(\Omega)} + C_{3.5})^{-1}$ is used to ensure that $\mathcal{S}_{i,\varepsilon}[v]$ admits a modulus of continuity that is uniform with respect to v . Next, we define a smooth approximation of the arg max operator as

$$(3.6) \quad a_\varepsilon(s) := \frac{\sigma_2 - \sigma_1}{2} \left(1 + \tanh \frac{s}{\varepsilon} \right) + \sigma_1.$$

Finally, the regularized coefficient is defined by

$$(3.7) \quad a_{i,\varepsilon}[v](\mathbf{x}) := a_\varepsilon(\mathcal{S}_{i,\varepsilon}[v](\mathbf{x})), \quad \text{for a.e. } \mathbf{x} \in \Omega.$$

Hence,

$$a_{i,\varepsilon} : W_q \rightarrow C(\bar{\Omega})$$

defines a continuous operator.

Compactness of $a_{i,\varepsilon}$. We first show that the operator $\mathcal{S}_{i,\varepsilon} : W_q \rightarrow C(\bar{\Omega})$ is compact. Indeed, using Young's convolution inequality (A.2), we have

$$(3.8) \quad \begin{aligned} \|\mathcal{S}_{i,\varepsilon}[v]\|_{L_\infty(\Omega)} &\leq \left(\|v\|_{W_q^2(\Omega)} + C_{3.5}\right)^{-1} \|\eta_\varepsilon * \partial_{x_i x_i} \tilde{v}\|_{L_\infty(\mathbb{R}^d)} \\ &\leq \left(\|v\|_{W_q^2(\Omega)} + C_{3.5}\right)^{-1} \|\eta_\varepsilon\|_{L_{\frac{q}{q-1}}(\mathbb{R}^d)} \|\tilde{v}\|_{W_q^2(\mathbb{R}^d)} \leq C_\varepsilon \frac{\|v\|_{W_q^2(\Omega)}}{\left(\|v\|_{W_q^2(\Omega)} + C_{3.5}\right)} \leq C_\varepsilon, \end{aligned}$$

with $C_\varepsilon \sim \varepsilon^{-d/q}$.

Moreover, for any multi-index $\mathbf{k} \in \mathbb{N}^d$,

$$\partial^{\mathbf{k}}(\eta_\varepsilon * \partial_{x_i x_i} \tilde{v}) = (\partial^{\mathbf{k}} \eta_\varepsilon) * \partial_{x_i x_i} \tilde{v},$$

so that

$$(3.9) \quad \begin{aligned} &\left(\|v\|_{W_q^2(\Omega)} + C_{3.5}\right)^{-1} \|\partial^{\mathbf{k}}(\eta_\varepsilon * \partial_{x_i x_i} \tilde{v})\|_{L_\infty(\mathbb{R}^d)} \\ &\leq \left(\|v\|_{W_q^2(\Omega)} + C_{3.5}\right)^{-1} \|\partial^{\mathbf{k}} \eta_\varepsilon\|_{L_{\frac{q}{q-1}}(\mathbb{R}^d)} \|\tilde{v}\|_{W_q^2(\mathbb{R}^d)} \\ &\leq \frac{C_\varepsilon}{\varepsilon^{|\mathbf{k}|}} \frac{\|v\|_{W_q^2(\Omega)}}{\left(\|v\|_{W_q^2(\Omega)} + C_{3.5}\right)} \leq \frac{C_\varepsilon}{\varepsilon^{|\mathbf{k}|}}. \end{aligned}$$

with $|\mathbf{k}| = |\mathbf{k}|_{l_\infty}$. If $\{v_m\}_m \subset W_q$ is bounded, $\|v_m\|_{W_q^2(\Omega)} \leq C$, then the sequence $\{\eta_\varepsilon * \partial_{x_i x_i} \tilde{v}_m\}_m \subset C^\infty(\mathbb{R}^d)$ is

- uniformly bounded in $L_\infty(\mathbb{R}^d)$,
- equicontinuous on compact subset $\bar{\Omega}$.

By the Arzelà–Ascoli theorem A.3, $\mathcal{S}_{i,\varepsilon}[v_m]$ has a uniformly convergent subsequence in $C(\bar{\Omega})$. Therefore, for each fixed $\varepsilon > 0$, the operator $\mathcal{S}_{i,\varepsilon} : W_q \rightarrow C(\bar{\Omega})$ is compact. Since a_ε is smooth, the composition $a_{i,\varepsilon} = a_\varepsilon \circ \mathcal{S}_{i,\varepsilon}$ is compact.

Pointwise convergence. For any $v \in W_q$,

$$(3.10) \quad \begin{aligned} \lim_{\varepsilon \rightarrow 0} \mathcal{S}_{i,\varepsilon}[v](\mathbf{x}) &= \left(\|v\|_{W_q^2(\Omega)} + C_{3.5}\right)^{-1} \lim_{\varepsilon \rightarrow 0} (\eta_\varepsilon * \partial_{x_i x_i} v)(\mathbf{x}) \\ &= \frac{\partial_{x_i x_i} v(\mathbf{x})}{\left(\|v\|_{W_q^2(\Omega)} + C_{3.5}\right)} \quad \text{a.e. in } \Omega. \end{aligned}$$

Hence,

$$(3.11) \quad \lim_{\varepsilon \rightarrow 0} \text{dist} \left(a_{i,\varepsilon}[v](\mathbf{x}), \left\{ \arg \max_{\alpha \in [\sigma_1, \sigma_2]} [\alpha \partial_{x_i x_i} v(\mathbf{x})] \right\} \right) = 0, \quad \text{a.e. in } \Omega.$$

Therefore, pointwise a.e. convergence is guaranteed.

Uniform modulus of continuity. Since $a_{\varepsilon_2}(s)$ is smooth with derivative

$$a'_\varepsilon(s) = \frac{\sigma_2 - \sigma_1}{2\varepsilon} \operatorname{sech}^2\left(\frac{s}{\varepsilon}\right),$$

we have

$$(3.12) \quad |a'_\varepsilon(s)| \leq \frac{\sigma_2 - \sigma_1}{2\varepsilon} \quad \forall s \in \mathbb{R}.$$

On the other hand, the upper bound (3.9) implies that for any $v \in W_q$,

$$(3.13) \quad \|\nabla \mathcal{S}_{i,\varepsilon}[v]\|_{L_\infty(\Omega)} \leq \frac{C_\varepsilon}{\varepsilon}.$$

Therefore

$$(3.14) \quad |\nabla a_{i,\varepsilon}[v](\mathbf{x})| = |a'_\varepsilon(\mathcal{S}_{i,\varepsilon}[v](\mathbf{x})) \nabla \mathcal{S}_{i,\varepsilon}[v](\mathbf{x})| \leq \frac{C_\varepsilon(\sigma_2 - \sigma_1)}{2\varepsilon^2} \quad \text{a.e. in } \Omega.$$

This bound is independent of v . Now, for any $\mathbf{x}, \mathbf{y} \in \Omega$,

$$(3.15) \quad |a_{i,\varepsilon}[v](\mathbf{x}) - a_{i,\varepsilon}[v](\mathbf{y})| \leq \|\nabla a_{i,\varepsilon}[v]\|_{L_\infty(\Omega)} |\mathbf{x} - \mathbf{y}| \leq \frac{C_\varepsilon(\sigma_2 - \sigma_1)}{2\varepsilon^2} |\mathbf{x} - \mathbf{y}|.$$

Thus the modulus of continuity $\omega[a_{i,\varepsilon}[v]](r) := \frac{C_\varepsilon(\sigma_2 - \sigma_1)}{2\varepsilon^2} r$ is uniform in v .

Preservation of the lower and upper bounds. Since

$$\sigma_1 \leq a_\varepsilon(s) \leq \sigma_2, \quad \forall s \in \mathbb{R},$$

it follows that for any $v \in W_q$

$$(3.16) \quad \sigma_1 \leq a_{i,\varepsilon}[v](\mathbf{x}) \leq \sigma_2, \quad \text{for a.e. } \mathbf{x} \in \Omega.$$

We present below the key tools, in the form of the following theorems, required to establish the existence of solutions to the regularized system (3.1).

Theorem 3.2 (well-posedness of a linear equation with continuous coefficients [3, Theorem 9.15 and Lemma 9.17]). *Let the diffusion coefficients $\mathbf{a} := (a_1, \dots, a_d)$ belong to $(C(\bar{\Omega}))^d$, and assume that*

$$\sigma_1 \leq a_i(\mathbf{x}) \leq \sigma_2 \quad \text{for a.e. } \mathbf{x} \in \Omega, \quad i = 1, \dots, d.$$

Let $c \in L_\infty(\Omega)$ with $c \geq 0$. If $f \in L_q(\Omega)$ and $g \in W_q^2(\Omega)$ with $1 < q < \infty$, then the equation

$$(3.17) \quad \sum_{i=1}^d a_i \partial_{x_i x_i} u - c u = f \quad \text{in } \Omega, \quad u = g \quad \text{on } \partial\Omega,$$

admits a unique solution $u \in W_q^2(\Omega)$. Moreover, there exists a constant $C_{3.18}(\Omega, q, \sigma_1, \sigma_2, \|c\|_{L_\infty(\Omega)}, \boldsymbol{\omega}[\mathbf{a}]) > 0$ such that the following estimate holds:

$$(3.18) \quad \|u\|_{W_q^2(\Omega)} \leq C_{3.18} \left(\|f\|_{L_q(\Omega)} + \|g\|_{W_q^2(\Omega)} \right).$$

Lemma 3.3 (Stability with respect to the diffusion coefficients). *Let $\bar{\mathbf{a}} = (\bar{a}_1, \dots, \bar{a}_d)$ and $\tilde{\mathbf{a}} = (\tilde{a}_1, \dots, \tilde{a}_d)$ belong to $(C(\bar{\Omega}))^d$. Let $c \in L_\infty(\Omega)$ with $c \geq 0$ a.e. in Ω . Assume that*

$$\sigma_1 \leq \bar{a}_i(\mathbf{x}), \quad \tilde{a}_i(\mathbf{x}) \leq \sigma_2 \quad \text{for a.e. } \mathbf{x} \in \Omega, \quad i = 1, \dots, d.$$

Let $\bar{u}, \tilde{u} \in W_q$ with $1 < q < \infty$, satisfy

$$(3.19) \quad \sum_{i=1}^d \bar{a}_i \partial_{x_i x_i} \bar{u} - c \bar{u} = f, \quad \sum_{i=1}^d \tilde{a}_i \partial_{x_i x_i} \tilde{u} - c \tilde{u} = f \quad \text{in } \Omega,$$

where $f \in L_q(\Omega)$. Then there exists a constant

$$C_{3.20}(\Omega, q, \sigma_1, \sigma_2, \|c\|_{L_\infty(\Omega)}, \|f\|_{L_q(\Omega)}, \omega[\bar{\mathbf{a}}], \omega[\tilde{\mathbf{a}}]) > 0$$

such that

$$(3.20) \quad \|\bar{u} - \tilde{u}\|_{W_q^2(\Omega)} \leq C_{3.20} \|\bar{\mathbf{a}} - \tilde{\mathbf{a}}\|_{L_\infty(\Omega)}.$$

Proof. Subtracting the two equations in (3.19), we obtain

$$\sum_{i=1}^d \bar{a}_i \partial_{x_i x_i} (\bar{u} - \tilde{u}) - c(\bar{u} - \tilde{u}) = \sum_{i=1}^d (\tilde{a}_i - \bar{a}_i) \partial_{x_i x_i} \tilde{u} \quad \text{in } \Omega.$$

Applying the stability estimate from Lemma 3.2 gives

$$\begin{aligned} & \|\bar{u} - \tilde{u}\|_{W_q^2(\Omega)} \\ & \leq C_{3.18} \left(\Omega, q, \sigma_1, \sigma_2, \|c\|_{L_\infty(\Omega)}, \omega[\bar{\mathbf{a}}] \right) \left\| \sum_{i=1}^d (\tilde{a}_i - \bar{a}_i) \partial_{x_i x_i} \tilde{u} \right\|_{L_q(\Omega)} \\ & \leq C_{3.18} \left(\Omega, q, \sigma_1, \sigma_2, \|c\|_{L_\infty(\Omega)}, \omega[\bar{\mathbf{a}}] \right) \|\bar{\mathbf{a}} - \tilde{\mathbf{a}}\|_{L_\infty(\Omega)} \|\tilde{u}\|_{W_q^2(\Omega)} \\ & \leq C_{3.18} \left(\Omega, q, \sigma_1, \sigma_2, \|c\|_{L_\infty(\Omega)}, \omega[\bar{\mathbf{a}}] \right) C_{3.18} \left(\Omega, q, \sigma_1, \sigma_2, \|c\|_{L_\infty(\Omega)}, \omega[\tilde{\mathbf{a}}] \right) \|f\|_{L_q(\Omega)} \|\bar{\mathbf{a}} - \tilde{\mathbf{a}}\|_{L_\infty(\Omega)}. \end{aligned}$$

The desired estimate follows by absorbing the dependence on the constants $C_{3.18}$ and on $\|f\|_{L_q(\Omega)}$ into $C_{3.20}$. $\square$

Theorem 3.4 (Schaefer's fixed point theorem [2, Section 9.2.2., Theorem 4]). *Let X be a Banach space and $\Phi : X \rightarrow X$ be a continuous and compact operator, such that the set*

$$(3.21) \quad \{x \in X \mid x = \eta \Phi(x) \text{ for some } \eta \in [0, 1]\},$$

is bounded. Then Φ has a fixed point in X ; ($\exists x \in X : \Phi(x) = x$).

Lemma 3.5 (existence of strong solution for the regularized system). *For any fixed $\varepsilon > 0$, there exists a solution*

$$\mathbf{u}_\varepsilon = (u_{1,\varepsilon}, \dots, u_{d,\varepsilon}) \in (W_q)^d$$

to the regularized system (3.1).

Proof. For any fixed $\varepsilon > 0$, define the operator $\Phi_\varepsilon : (W_q)^d \rightarrow (W_q)^d$ by

$$(3.22) \quad \mathbf{v} := (v_1, \dots, v_d) \mapsto \Phi_\varepsilon(v_1, \dots, v_d) := (u_1, \dots, u_d),$$

where $(u_1, \dots, u_d) \in (W_q)^d$ is the unique solution of the linear decoupled equations

$$(3.23) \quad \sum_{i=1}^d a_{i,\varepsilon} [v_i] \partial_{x_i x_i} u_j - c_j u_j = f_j, \quad \text{in } \Omega, \quad j = 1, \dots, d.$$

By Theorem 3.2, for any $(v_1, \dots, v_d) \in (W_q)^d$, the decoupled system (3.23) has a unique solution $\mathbf{u} := (u_1, \dots, u_d)$, and it satisfies the a priori estimate

$$(3.24) \quad \|\mathbf{u}\|_{W_q^2(\Omega)} \leq C_{3.24}(\Omega, q, \sigma_1, \sigma_2, \|\mathbf{c}\|_{L_\infty(\Omega)}, \|\mathbf{f}\|_{L_q(\Omega)}, \omega[\mathbf{a}_\varepsilon[\mathbf{v}]]).$$

Since the modulus of continuity $\omega[\mathbf{a}_\varepsilon[\mathbf{v}]]$ is independent of $\mathbf{v}$, it follows that

$$(3.25) \quad \|\mathbf{u}\|_{W_q^2(\Omega)} \leq C_{3.25}(\Omega, q, \sigma_1, \sigma_2, \|\mathbf{c}\|_{L_\infty(\Omega)}, \|\mathbf{f}\|_{L_q(\Omega)}, \varepsilon).$$

We now show that Φ_ε is compact. Let $\{(v_{1,m}, \dots, v_{d,m})\}_m \subset (W_q)^d$ be a bounded sequence ($\|\mathbf{v}_m\|_{W_p^2(\Omega)} \leq C$), and denote

$$(u_{1,m}, \dots, u_{d,m}) := \Phi_\varepsilon(v_{1,m}, \dots, v_{d,m}).$$

Since the operators $a_{i,\varepsilon} : W_q \rightarrow C(\bar{\Omega})$ are compact, there exists a subsequence $\{(v_{1,m_n}, \dots, v_{d,m_n})\}_{m_n}$ and $(v_1, \dots, v_d) \in (W_q)^d$ such that

$$(3.26) \quad \lim_{n \rightarrow \infty} \left\| (a_{1,\varepsilon}[v_{1,m_n}], \dots, a_{d,\varepsilon}[v_{d,m_n}]) - (a_{1,\varepsilon}[v_1], \dots, a_{d,\varepsilon}[v_d]) \right\|_{L_\infty(\Omega)} = 0.$$

Then, by the stability estimate (3.20) there exists

$$C_{3.27}(\Omega, q, \mathbf{f}, \|\mathbf{c}\|_{L_\infty(\Omega)}, \sigma_1, \sigma_2, \varepsilon) > 0$$

such that

$$(3.27) \quad \begin{aligned} & \| (u_{1,m_n}, \dots, u_{d,m_n}) - (u_1, \dots, u_d) \|_{W_q^2(\Omega)} \\ & \leq C_{3.27} \left\| (a_{1,\varepsilon}[v_{1,m_n}], \dots, a_{d,\varepsilon}[v_{d,m_n}]) - (a_{1,\varepsilon}[v_1], \dots, a_{d,\varepsilon}[v_d]) \right\|_{L_\infty(\Omega)}, \end{aligned}$$

where $(u_1, \dots, u_d) = \Phi_\varepsilon(v_1, \dots, v_d)$. This shows that $\{(u_{1,m}, \dots, u_{d,m})\}_m$ has a convergent subsequence in $(W_q)^d$, establishing the compactness of Φ_ε .

The estimate (3.25) implies that the set (3.21) is uniformly bounded (for fixed ε).

Finally, the Schaefer's fixed point theorem guarantees the existence of a fixed point

$$(u_{1,\varepsilon}, \dots, u_{d,\varepsilon}) \in (W_q)^d \quad \text{such that} \quad \Phi_\varepsilon(u_{1,\varepsilon}, \dots, u_{d,\varepsilon}) = (u_{1,\varepsilon}, \dots, u_{d,\varepsilon}),$$

i.e., $(u_{1,\varepsilon}, \dots, u_{d,\varepsilon})$ is a solution of the regularized system (3.1). $\square$

We next analyze the behavior of the family of solutions to the regularized system and use this to define a solution to the original coupled HJB system (3.1). This requires uniform estimates for the family of regularized solutions, which are obtained by imposing the Cordes condition on the diffusion coefficients.

Assumption 1 (Cordes condition). *We further suppose that the coefficients a_i for $i = 1, \dots, d$ satisfy the Cordes condition: there exists $\lambda > 0$ such that for any $\mathbf{v} := (v_1, \dots, v_d) \in (W_q)^d$,*

$$(3.28) \quad \frac{\sum_{i=1}^d (a_i [v_i](\mathbf{x}))^2}{\left(\sum_{i=1}^d a_i [v_i](\mathbf{x})\right)^2} \leq \frac{1}{d-1+\lambda} \quad \text{for a.e. } \mathbf{x} \in \Omega.$$

Consequently, the regularized coefficients $a_{i,\varepsilon} [v_i]$ satisfy the Cordes condition as well.

Now that the diffusion coefficients satisfy the Cordes condition, the results of [4] apply with $k = 2$. In particular, the following a priori estimates for solutions to the regularized system (3.1) in $(W_p)^d$ follow from the estimates collected in the Appendix B.1, some of which are taken from the literature and some established therein.

Remark 3.6. We point out that, since

$$\mathbf{u}_\varepsilon \in (W_p)^d,$$

is a solution to the regularized system (3.1) with $p > d/2$, the Sobolev embedding theorem [2, Section 5.6.3, Theorem 6] implies that

$$\mathbf{u}_\varepsilon \in (C(\overline{\Omega}))^d.$$

Lemma 3.7 (ABP estimate [4, Corollary 4.2]). *Let*

$$\mathbf{u}_\varepsilon = (u_{1,\varepsilon}, \dots, u_{d,\varepsilon}) \in (W_p)^d$$

be the solution to the regularized system (3.1). Then

$$(3.29) \quad \|\mathbf{u}_\varepsilon\|_{L_\infty(\Omega)} \leq C_{3.29} \|\mathbf{f}\|_{L_p(\Omega)}, \quad \forall \varepsilon > 0,$$

where $C_{3.29}(\Omega, p, \sigma_1, \sigma_2, \lambda, \|\mathbf{c}\|_{L_\infty(\Omega)}) > 0$ is independent of the modulus of continuity of the coefficients.

Proof. See Lemma B.1 in the Appendix. $\square$

Lemma 3.8 (Hölder continuity). *There exist*

$$\gamma(d, p, \sigma_2, \lambda), C_{3.30}(\Omega, p, \sigma_1, \sigma_2, \lambda, \|\mathbf{c}\|_{L_\infty}, \|\mathbf{f}\|_{L_p}, \|\mathbf{u}_\varepsilon\|_{L_\infty}) > 0$$

such that the solution $\mathbf{u}_\varepsilon = (u_{1,\varepsilon}, \dots, u_{d,\varepsilon}) \in (W_p)^d$ of the regularized system (3.1) belongs to $(C^{0,\gamma}(\overline{\Omega}))^d$ and satisfies

$$(3.30) \quad \|\mathbf{u}_\varepsilon\|_{C^{0,\gamma}(\Omega)} \leq C_{3.30}, \quad \forall \varepsilon > 0.$$

Proof. The result follows from Lemma B.5 proved in the Appendix. $\square$

Remark 3.9. We emphasize that the Hölder constant $C_{3.30}$ is independent of the modulus of continuity of the diffusion coefficients, $\omega[a_\varepsilon[\mathbf{u}_\varepsilon]]$.

Remark 3.10. Combining Lemma 3.7 and Lemma 3.8, we conclude that the Hölder continuity constant $C_{3.30}$ can be chosen independently of $\|\mathbf{u}_\varepsilon\|_{L_\infty(\Omega)}$.

Remark 3.11 (equicontinuity). The Hölder continuity estimate (3.30) implies that the family $\{\mathbf{u}_\varepsilon\}_\varepsilon$ of solutions to the system (3.1) is equicontinuous.

Proof. Lemma 3.8 implies that there exist $\gamma, C_{3.30} > 0$ such that

$$(3.31) \quad \|\mathbf{u}_\varepsilon\|_{C^{0,\gamma}(\bar{\Omega})} := \|\mathbf{u}_\varepsilon\|_{L^\infty(\Omega)} + [\mathbf{u}_\varepsilon]_{C^{0,\gamma}(\Omega)} \leq C_{3.30}, \quad \forall \varepsilon > 0.$$

Since

$$(3.32) \quad |\mathbf{u}_\varepsilon(\mathbf{x}) - \mathbf{u}_\varepsilon(\mathbf{y})| \leq [\mathbf{u}_\varepsilon]_{C^{0,\gamma}(\Omega)} |\mathbf{x} - \mathbf{y}|^\gamma \quad \forall \varepsilon > 0, (\mathbf{x}, \mathbf{y}) \in \bar{\Omega} \times \bar{\Omega},$$

for any $\zeta > 0$, by choosing $\delta = \left(\frac{\zeta}{C_{3.30}}\right)^{\frac{1}{\gamma}}$, for all $\varepsilon > 0$ and all $\mathbf{x}, \mathbf{y}$ with $|\mathbf{x} - \mathbf{y}| < \delta$ we have

$$(3.33) \quad |\mathbf{u}_\varepsilon(\mathbf{x}) - \mathbf{u}_\varepsilon(\mathbf{y})| \leq \zeta. \quad \square$$

4. STATIONARY COUPLED HJB SYSTEM

Now that the existence of solutions to the regularized system has been established, we aim to relate these solutions to a solution of the original coupled HJB system (2.2).

In the following, we adopt a notion of solution that was previously introduced for linear equations with measurable diffusion coefficients in [9, 1]. We extend this concept to the present fully nonlinear coupled system.

Definition 4.1 (weak solution or limit solution). A function $\mathbf{u} := (u_1, \dots, u_d) \in (C(\bar{\Omega}))^d$ is called a weak solution of the HJB system (2.2) if there exists a subsequence $\{\mathbf{u}_\varepsilon\}_\varepsilon$ of solutions to the regularized system (3.1) such that

$$\lim_{\varepsilon \rightarrow 0} \|\mathbf{u}_\varepsilon - \mathbf{u}\|_{C(\bar{\Omega})} = 0.$$

In the regularized fully nonlinear setting, one only has control of $A_\varepsilon[\mathbf{v}]$ for a fixed (vector-valued) function $\mathbf{v}$, and not of the full nonlinear coefficients $A_\varepsilon[\mathbf{u}_\varepsilon]$. Consequently, the passage to the limit in the nonlinear coefficients, namely the convergence of $A_\varepsilon[\mathbf{u}_\varepsilon]$ to $A[\mathbf{u}]$, is delicate. For this reason, it is not clear a priori that every strong solution of the original system is a limit solution. Likewise, even when a limit solution enjoys sufficient regularity, it is not immediate that it is a strong solution of the original system.

Lemma 4.2 (existence of weak solution for the fully coupled HJB system). *The fully coupled system (2.2) has a weak solution.*

Proof. Let $\{\mathbf{u}_\varepsilon\}_\varepsilon$ be the family of solutions to the regularized system (3.1). By the a priori estimate (3.29), this family is uniformly bounded. Moreover, Remark 3.11 guarantees equicontinuity. Therefore, by the Arzelà–Ascoli theorem A.3, there exists a subsequence that converges uniformly in $(C(\bar{\Omega}))^d$ to a limit $\mathbf{u}$. Hence, $\mathbf{u}$ is a weak solution of the fully coupled system (2.2). $\square$

To obtain additional regularity of the solution, we introduce the exponent $\hat{p}$ defined by

$$(4.1) \quad \hat{p} := \begin{cases} 2, & \text{if } p = 2, \\ 2 + \varrho < p, & \text{if } p > 2, \end{cases} = \begin{cases} 2, & \text{if } d = 2, 3, \\ 2 + \varrho < p, & \text{if } d \geq 4, \end{cases}$$

for some sufficiently small $\varrho > 0$. With this choice, one can ensure that the following uniform a priori estimate holds.

Lemma 4.3 (uniform a priori $W_{\hat{p}}^2$ estimate for the regularized system). *Let*

$$\mathbf{u}_\varepsilon = (u_{1,\varepsilon}, \dots, u_{d,\varepsilon}) \in (W_p)^d$$

be the solution to the regularized system (3.1). Assume that the diffusion coefficients satisfy the Cordes condition (3.28) and that the domain Ω is convex. Then there exist $\varrho > 0$ and, consequently, an exponent $\hat{p}$ defined by (4.1), such that there exists a constant

$$C_{4.2}(\Omega, \sigma_1, \sigma_2, \lambda, p, \|c\|_{L_\infty}) > 0$$

with the property that the following uniform a priori estimate holds:

$$(4.2) \quad \|u_{j,\varepsilon}\|_{W_{\hat{p}}^2(\Omega)} \leq C_{4.2} \|f_j\|_{L_p(\Omega)}, \quad \forall \varepsilon > 0, \quad j = 1, \dots, d.$$

Proof. For each $j = 1, \dots, d$, the regularized equations read

$$(4.3) \quad \sum_{i=1}^d a_{i,\varepsilon} [u_{i,\varepsilon}] \partial_{x_i x_i} u_{j,\varepsilon} = f_j + c_j u_{j,\varepsilon} \quad \text{in } \Omega.$$

The Cordes condition yields

$$(4.4) \quad \|u_{j,\varepsilon}\|_{W_{\hat{p}}^2(\Omega)} \leq C_{4.4} \left(\|f_j\|_{L_{\hat{p}}(\Omega)} + \|c_j u_{j,\varepsilon}\|_{L_{\hat{p}}(\Omega)} \right),$$

where $C_{4.4}(\Omega, \sigma_1, \sigma_2) > 0$ (see [7, Theorem 1.2.3]). By Lemma 3.7, we have

$$\|c_j u_{j,\varepsilon}\|_{L_{\hat{p}}(\Omega)} \leq C(\Omega) C_{3.29} \|c_j\|_{L_\infty(\Omega)} \|f_j\|_{L_p(\Omega)}.$$

Substituting this estimate into (4.4) yields the desired result. $\square$

Remark 4.4 ($W_{\hat{p}}^2$ -regularity of a weak solution). Lemma 4.3 implies that the family

$$\{\mathbf{u}_\varepsilon := (u_{1,\varepsilon}, \dots, u_{d,\varepsilon})\}_\varepsilon,$$

of solutions to the regularized system (3.1) is uniformly bounded in $W_{\hat{p}}^2(\Omega)$. Consequently, there exists a subsequence (not relabeled) and a function $\bar{\mathbf{u}} \in (W_{\hat{p}}^2(\Omega))^d$ such that

$$\mathbf{u}_\varepsilon \rightharpoonup \bar{\mathbf{u}} \quad \text{weakly in } W_{\hat{p}}^2(\Omega).$$

By uniqueness of the limit, we have $\bar{\mathbf{u}} = \mathbf{u}$, where $\mathbf{u}$ denotes the weak solution of (2.2). It follows that $\mathbf{u} \in (W_{\hat{p}}^2(\Omega))^d$.

In the following lemma, we provide a sufficient condition that allows passage to the limit in the equation, thereby ensuring that a weak solution is in fact a strong solution.

4.1. A Cordes condition approach for $d \leq 4$. When considering the problem in dimensions $d \leq 4$ with vanishing zero-order term, i.e. $\mathbf{c} = \mathbf{0}$, one may alternatively assume that $\Omega \subset \mathbb{R}^d$ is a convex domain without requiring $C^{1,1}$ regularity. In this setting, when dealing with the regularized system, one can employ the Cordes condition framework instead of Theorem 3.2. This yields a uniform a priori estimate for the solution, independent of the regularization parameter.

The Cordes condition framework is particularly well suited to the case $d \leq 4$. Indeed, the following theorem ensures the well-posedness of solutions in $W_p^2(\Omega) \cap \dot{W}_p^1(\Omega)$, for

any exponent p satisfying the constraints in (2.1). As a consequence, both the ABP estimate (3.29) and the Hölder continuity estimate (3.30) remain valid. The latter follows from the continuous embedding of $W_{\hat{p}}^2(\Omega)$ into $C^{0,\alpha}(\bar{\Omega})$ (see [2, Section 5.6.3, Theorem 6]), together with the uniform estimate provided by Theorem 4.5.

In contrast, for dimensions $d \geq 5$, the same theorem no longer guarantees the existence of solutions in $W_p^2(\Omega) \cap \dot{W}_p^1(\Omega)$ for admissible exponents p . This loss of regularity leads to significant difficulties in continuing the analysis. (The point is that for $d \leq 4$, the exponents p and $\hat{p}$, can be used interchangeably. In contrast, for higher dimensions, such an interchangeability is no longer valid.)

Theorem 4.5 (well-posedness of a linear equation with coefficients satisfying Cordes condition and $d \leq 4$ [7, Theorem 1.2.3]). *Let Ω be a bounded, convex domain and let the coefficients $\mathbf{a} := (a_1, \dots, a_d)$ satisfy the Cordes condition. Assume further that*

$$\sigma_1 \leq a_i(\mathbf{x}) \leq \sigma_2 \quad \text{for a.e. } \mathbf{x} \in \Omega, \quad i = 1, \dots, d.$$

If $f \in L_{\hat{p}}(\Omega)$, then the equation

$$(4.5) \quad \sum_{i=1}^d a_i \partial_{x_i x_i} u = f \quad \text{in } \Omega, \quad u_j = 0 \quad \text{on } \partial\Omega,$$

admits a unique solution

$$u \in W_{\hat{p}}^2(\Omega) \cap \dot{W}_{\hat{p}}^1(\Omega),$$

where $\hat{p}$ is as defined in (4.1). Moreover, there exists $C_{4.6}(\Omega, d, \hat{p}, \sigma_1, \sigma_2, \lambda) > 0$ such that the estimate

$$(4.6) \quad \|u\|_{W_{\hat{p}}^2(\Omega)} \leq C_{4.6} \|f\|_{L_{\hat{p}}(\Omega)},$$

holds.

The existence of a weak solution in this setting then follows by the same argument as in the general case.

5. TIME-DEPENDENT COUPLED HJB SYSTEM

Here, we consider a time-dependent coupled HJB system and adapt the arguments developed above to this setting.

Let Ω be as before, and fix a final time $T > 0$. Define the space–time cylinder

$$Q_T := \Omega \times (0, T),$$

and denote its lateral boundary by

$$S_T := \partial\Omega \times (0, T).$$

For $1 \leq q, r < \infty$, we define $L_{q,r}(Q_T)$ as the Banach space of all measurable functions v on Q_T such that the norm

$$\|v\|_{L_{q,r}(Q_T)} := \left(\int_0^T \left(\int_{\Omega} |v(\mathbf{x}, t)|^q dx \right)^{\frac{r}{q}} dt \right)^{\frac{1}{r}}.$$

is finite. Define the space when q or r be ∞ as well. When $q = r$, we simply write

$$L_q(Q_T) := L_{q,q}(Q_T), \quad \|v\|_{L_q(Q_T)} := \|v\|_{L_{q,q}(Q_T)}.$$

Furthermore, for integers $m, n \geq 0$, and $1 \leq q < \infty$, we denote by $W_q^{m,n}(Q_T)$ the anisotropic Sobolev space consisting of functions on Q_T whose weak derivatives up to order m with respect to the spatial variables $\mathbf{x}$ and up to order n with respect to the time variable t belong to $L_q(Q_T)$. In particular, $v \in W_q^{2,1}(Q_T)$ if

$$v, \partial_t v, \partial_{x_i} v, \partial_{x_i x_j} v \in L_q(Q_T) \quad \text{for } i, j = 1, \dots, d.$$

Remark 5.1. In the theory of parabolic partial differential equations, it is customary to use the *parabolic metric*

$$|(\mathbf{x}, t) - (\mathbf{y}, s)| := |\mathbf{x} - \mathbf{y}| + |t - s|^{1/2},$$

instead of the Euclidean metric in $\mathbb{R}^{d+1}$.

Theorem 5.2 (well-posedness of a linear time-dependent equation with continuous coefficients [6, Chapter IV, Theorem 9.1]). *Let $\frac{3}{2} < q < \infty$. Suppose the coefficients $\mathbf{a} := (a_1, \dots, a_d)$ belong to $(C(\bar{Q}_T))^d$, and assume that*

$$\sigma_1 \leq a_i(\mathbf{x}, t) \leq \sigma_2 \quad \text{for a.e. } (\mathbf{x}, t) \in Q_T, \quad i = 1, \dots, d.$$

Let $c \in L_\infty(Q_T)$ with $c \geq 0$. If $f \in L_q(Q_T)$, $u_0 \in W_q^{2-2/q}(\Omega)$, and $g \in W_q^{2-1/q, 2-1/2q}(S_T)$, and the compatibility condition

$$u_0 = g(\cdot, 0) \quad \text{on } \partial\Omega,$$

holds, then the equation

$$(5.1) \quad \sum_{i=1}^d \partial_t u - a_i \partial_{x_i x_i} u + c u = f \quad \text{in } Q_T, \quad u|_{t=0} = u_0, \quad u|_{S_T} = g,$$

admits a unique solution $u \in W_q^{2,1}(Q_T)$. Moreover, there exists a constant $C_{5.2}(\Omega, T, q, \sigma_1, \sigma_2, \|\hat{c}\|_{L_\infty(Q_T)}, \omega[\hat{\mathbf{a}}]) > 0$ such that the following estimate holds:

$$(5.2) \quad \|u\|_{W_q^{2,1}(Q_T)} \leq C_{5.2} \left(\|f\|_{L_q(Q_T)} + \|g\|_{W_q^{2-\frac{1}{q}, 1-\frac{1}{2q}}(S_T)} + \|u_0\|_{W_q^{2-\frac{2}{q}}(\Omega)} \right).$$

We define

$$(5.3) \quad W_{T, \bar{p}} := \left\{ v \in W_{\bar{p}}^{2,1}(Q_T) : v(\mathbf{x}, t) = 0 \text{ on } S_T, \quad v(\mathbf{x}, 0) = 0 \text{ on } \Omega \right\},$$

where the exponent $\bar{p}$ is assumed to satisfy

$$\frac{d+2}{2} < \bar{p} < \infty.$$

We now consider the time dependent fully coupled system of HJB equations, given by

$$(5.4) \quad \begin{cases} \sum_{i=1}^d \partial_t u_j - a_i[u_i] \partial_{x_i x_i} u_j + c_j u_j = f_j, & \text{in } Q_T, \\ u_j|_{t=0} = 0, & \text{in } \Omega, \\ u_j = 0, & \text{in } S_T, \end{cases} \quad j = 1, \dots, d,$$

for any $\mathbf{f} \in (L_{\bar{p}}(Q_T))^d$. Here, for any $v \in W_{\bar{p}}^{2,1}(Q_T)$ and $i = 1, \dots, d$, the nonlinear coefficients are defined by

$$a_i[v](\mathbf{x}, t) \in \left\{ \arg \sup_{\alpha \in [\sigma_1, \sigma_2]} \{ \alpha \partial_{x_i x_i} v(\mathbf{x}, t) \} \right\}.$$

For the same reasons as in the stationary fully coupled HJB system, we also consider, in the time-dependent setting, a corresponding regularized version. The time-dependent regularized diffusion coefficients $a_{i,\varepsilon} : W_{T,\bar{p}} \rightarrow C(Q_T)$ are introduced with analogous properties. Applying an analysis similar to that of the stationary case, with Ω replaced by Q_T , p replaced by $\bar{p}$, and W_p replaced by $W_{T,\bar{p}}$, yields the existence of a corresponding weak solution in the time-dependent setting. The regularity of this weak solution can be established by arguments similar to those used in the stationary case [5].

Appendix A. Preliminaries and Auxiliary Tools.

Definition A.1. Let $v : \Omega \rightarrow \mathbb{R}$. The modulus of continuity of v is the function $\omega[v] : [0, \infty) \rightarrow [0, \infty)$ defined by

$$\omega[v](r) := \sup_{\substack{\mathbf{x}, \mathbf{y} \in \Omega \\ |\mathbf{x} - \mathbf{y}| \leq r}} |v(\mathbf{x}) - v(\mathbf{y})|, \quad r \geq 0.$$

Note that v is uniformly continuous in Ω if and only if $\omega[v](r) \rightarrow 0$ as $r \rightarrow 0$. For a vector-valued function $\mathbf{v} : \Omega \rightarrow \mathbb{R}^d$, with $\mathbf{v} = (v_1, \dots, v_d)$, the modulus of continuity is defined componentwise by

$$\omega[\mathbf{v}](r) := (\omega[v_1](r), \dots, \omega[v_d](r)).$$

Lemma A.2 (Young's convolution inequality). *Suppose f is in the Lebesgue space $L_s(\mathbb{R}^d)$ and g is in $L_q(\mathbb{R}^d)$ and*

$$\frac{1}{s} + \frac{1}{q} = \frac{1}{r} + 1$$

with $1 \leq s, q, r \leq \infty$. Then

$$(A.1) \quad \|f * g\|_{L_r(\mathbb{R}^d)} \leq \|f\|_{L_s(\mathbb{R}^d)} \|g\|_{L_q(\mathbb{R}^d)}.$$

Theorem A.3 (Arzelà–Ascoli theorem [8, Theorem 7.25]). *Let X be a compact metric space (we can take it as a compact subset of $\mathbb{R}^d$), and also let $C(X)$ denote the space of all continuous functions on X with values in the m -dimensional Euclidean space $\mathbb{R}^m$. If the sequence $\{v_n\}$ in $C(X)$ is uniformly bonded and uniformly equicontinuous then it has a uniformly convergent subsequence.*

Lemma A.4 (strong-weak continuity). *If $\{v_n\}$ be a sequence converges strongly to v ($v_n \rightarrow v$) and $\{w_n\}$ converges weakly to w ($w_n \rightharpoonup w$) in $L_2(\Omega)$, then*

$$\int_{\Omega} v_n w_n d\mathbf{x} \rightarrow \int_{\Omega} v w d\mathbf{x}.$$

Appendix B. A priori estimates. In the following, we assume that $\partial\Omega$ satisfies a uniform exterior sphere condition. In particular, this assumption holds whenever $\partial\Omega \in C^{1,1}$. We first establish the global Hölder continuity in the elliptic setting using the results of [4]. We then turn to the parabolic case using the corresponding results of [5].

B.1. Elliptic setting. Let

$$u \in W_p^2(\Omega) \cap C(\overline{\Omega}),$$

where we recall that the exponent p satisfies

$$p \geq 2, \quad \frac{d}{2} < p < \infty.$$

Assume that u satisfies

$$(B.1) \quad \mathcal{L}u := \sum_{i=1}^d a_i \partial_{x_i x_i} u - cu \geq -f \quad \text{in } \Omega, \quad u \leq 0 \quad \text{on } \partial\Omega,$$

where

$$f \in L_p(\Omega), \quad c \in L_{\infty}(\Omega), \quad c \geq 0,$$

and

$$\mathbf{A} = \text{diag}(a_1, \dots, a_d)$$

satisfies the ellipticity condition

$$(B.2) \quad \sigma_1 \leq a_i(\mathbf{x}) \leq \sigma_2, \quad \text{for a.e. } \mathbf{x} \in \Omega, \quad i = 1, \dots, d,$$

together with the Cordes condition: there exists $\lambda > 0$ such that

$$(B.3) \quad \frac{\sum_{i=1}^d a_i(\mathbf{x})^2}{\left(\sum_{i=1}^d a_i(\mathbf{x})\right)^2} \leq \frac{1}{d-1+\lambda} \quad \text{for a.e. } \mathbf{x} \in \Omega.$$

By definition,

$$\rho_m^*(\mathbf{A}) := \inf \left\{ \frac{\mathbf{A} \cdot \boldsymbol{\mu}}{d} : \boldsymbol{\mu} \in \Gamma_m, \rho_m(\boldsymbol{\mu}) \geq 1 \right\}.$$

The cones satisfy $\Gamma_k \subset \Gamma_l$ for $k > l$ (this is mentioned in the paper, the line after (1.9)). While Maclaurin inequality gives $\rho_k \leq \rho_l$. Hence

$$\{\boldsymbol{\mu} \in \Gamma_k : \rho_k(\boldsymbol{\mu}) \geq 1\} \subset \{\boldsymbol{\mu} \in \Gamma_l : \rho_l(\boldsymbol{\mu}) \geq 1\}.$$

Since ρ_k^* is an infimum of $\frac{\mathbf{A} \cdot \boldsymbol{\mu}}{d}$ over the left set, and ρ_l^* over the larger right set, the infimum over the smaller set is larger. So the monotonicity is:

$$\rho_1^*(\mathbf{A}) \leq \rho_2^*(\mathbf{A}) \leq \dots \leq \rho_d^*(\mathbf{A}).$$

In accordance with the notation in [4], we define

$$(B.4) \quad \rho_2^*(\mathbf{A}) := \frac{1}{\sqrt{d}} \left(\left(\sum_{i=1}^d a_i \right)^2 - (d-1) \sum_{i=1}^d a_i^2 \right)^{1/2}.$$

The ellipticity assumption together with the Cordes condition (B.3) satisfied by $\mathbf{A}$ implies

$$(B.5) \quad \rho_2^*(\mathbf{A}) \geq \lambda \sigma_1.$$

Therefore, if the diffusion matrix $\mathbf{A}$ satisfies the ellipticity condition together with the Cordes condition (B.3), then the results of [4] apply with $k = 2$. In particular, the function u satisfies the following estimates.

Lemma B.1 (ABP estimate [4, Corollary 4.2]). *Let $u \in W_p^2(\Omega) \cap C(\overline{\Omega})$ satisfy (B.1). Then we have the estimate*

$$(B.6) \quad \|u\|_{L_\infty(\Omega)} \leq C |\Omega|^{2/d-1/p} \left\| \frac{f}{\rho_2^*(\mathbf{A})} \right\|_{L_p(\Omega)},$$

for some $C(d, p, \sigma_1, \sigma_2, \lambda) > 0$.

Remark B.2. The estimate is originally stated for the supremum value of u . For continuous solutions, this is equivalent to an L_∞ upper bound. A full L_∞ bound on u requires applying the result to both u and $-u$. Therefore, [4, Corollary 4.2] provides an L_∞ estimate for the positive part of u ; applying it to u and $-u$ yields a full L_∞ bound.

Lemma B.3 (interior Hölder continuity [4, Theorem 4.3]). *Let $u \in W_p^2(\Omega)$ satisfy $\mathcal{L}u = f$ in $B_R(\mathbf{y}) \subset \Omega$. Then, for any concentric subball $B_{\sigma R}(\mathbf{y})$ with $0 < \sigma < 1$, the following oscillation estimate holds:*

$$(B.7) \quad \text{osc}_{B_{\sigma R}(\mathbf{y})} u \leq C \sigma^\alpha \left(\text{osc}_{B_R(\mathbf{y})} u + R^{2-d/p} \left\| \frac{f}{\rho_2^*(\mathbf{A})} \right\|_{L_p(B_R(\mathbf{y}))} \right),$$

for some $\alpha(d, \sigma_1, \sigma_2, \lambda), C(R, d, p, \sigma_1, \sigma_2, \lambda, \|c\|_{L_\infty}) > 0$. Consequently,

$$u \in C^{0,\alpha}(B_{\sigma R}(\mathbf{y})),$$

and

$$(B.8) \quad [u]_{C^{0,\alpha}(B_{\sigma R}(\mathbf{y}))} \leq C R^{-\alpha} \left(\text{osc}_{B_R(\mathbf{y})} u + R^{2-d/p} \left\| \frac{f}{\rho_2^*(\mathbf{A})} \right\|_{L_p(B_R(\mathbf{y}))} \right).$$

Remark B.4 (interior Hölder continuity). Lemma B.3 is an interior estimate: it assumes $B_R(\mathbf{y}) \subset \Omega$, and concludes Hölder continuity only on smaller concentric balls $B_{\sigma R}(\mathbf{y})$. Hence, it implies only local Hölder regularity:

$$(B.9) \quad u \in C_{\text{loc}}^{0,\alpha}(\Omega).$$

Now, by constructing a suitable boundary barrier and combining it with the interior Hölder estimate, we obtain global Hölder continuity $u \in C^{0,\gamma}(\overline{\Omega})$, as stated in the following lemma.

Lemma B.5 (global Hölder continuity). *Let $u \in W_p^2(\Omega) \cap C(\overline{\Omega})$ satisfy*

$$\mathcal{L}u = f \quad \text{in } \Omega, \quad u = 0 \quad \text{on } \partial\Omega.$$

Then there exist $\gamma(d, p, \sigma_1, \sigma_2, \lambda), C(\Omega, p, \sigma_1, \sigma_2, \lambda, \|c\|_{L^\infty}) > 0$ such that

$$(B.10) \quad \|u\|_{C^{0,\gamma}(\overline{\Omega})} \leq C \left(\|u\|_{L^\infty(\Omega)} + \left\| \frac{f}{\rho_2^*(\mathbf{A})} \right\|_{L^p(\Omega)} \right).$$

Proof. The interior Hölder regularity of u is already available from Lemma B.3, which ensures that

$$u \in C_{\text{loc}}^{0,\alpha}(\Omega)$$

for some $\alpha > 0$. Thus, it remains to establish Hölder continuity up to the boundary. Let $\mathbf{x}_0 \in \partial\Omega$. Since $\partial\Omega$ satisfies a uniform exterior sphere condition, there exists $R_0 > 0$ and a ball

$$B_{R_0}(\mathbf{z}) \subset \mathbb{R}^d \setminus \Omega, \quad \overline{B_{R_0}(\mathbf{z})} \cap \overline{\Omega} = \{\mathbf{x}_0\}.$$

Set $r(\mathbf{x}) := |\mathbf{x} - \mathbf{z}|$, so that $r(\mathbf{x}) \geq R_0$ for $\mathbf{x} \in \Omega$. For $\mu > 0$ large, define the barrier

$$(B.11) \quad \psi(\mathbf{x}) := R_0^{-\mu} - |\mathbf{x} - \mathbf{z}|^{-\mu}.$$

As in the standard computation, choosing μ sufficiently large yields

$$(B.12) \quad \mathcal{L}\psi \leq -C_{B.12} \quad \text{in } \Omega \cap B_{R_1}(\mathbf{x}_0),$$

for some $C_{B.12} > 0$. Now define

$$\Psi(\mathbf{x}) := \psi(\mathbf{x}) + \tau|\mathbf{x} - \mathbf{x}_0|^2,$$

with $\tau > 0$ small enough so that

$$\mathcal{L}\Psi \leq -\frac{C_{B.12}}{2}.$$

Fix $0 < R < R_1$ and set $D_R := \Omega \cap B_R(\mathbf{x}_0)$. Define

$$v := u - M_R \Psi, \quad M_R := \frac{\|u\|_{L^\infty(\Omega)}}{\tau R^2}.$$

By construction,

$$\mathcal{L}v \geq -|f| \quad \text{in } D_R, \quad v \leq 0 \quad \text{on } \partial D_R.$$

Hence, the ABP estimate applies and yields

$$\sup_{D_R} |v| \leq C R^{2-d/p} \left\| \frac{f}{\rho_2^*(\mathbf{A})} \right\|_{L_p(D_R)}.$$

Therefore,

$$|u(\mathbf{x})| \leq M_R \Psi(\mathbf{x}) + C R^{2-d/p} \left\| \frac{f}{\rho_2^*(\mathbf{A})} \right\|_{L_p(D_R)}.$$

Using the barrier estimate $\Psi(\mathbf{x}) \leq C|\mathbf{x} - \mathbf{x}_0|$, we obtain

$$|u(\mathbf{x})| \leq C \|u\|_{L^\infty(\Omega)} \frac{|\mathbf{x} - \mathbf{x}_0|}{R^2} + C R^{2-d/p} \left\| \frac{f}{\rho_2^*(\mathbf{A})} \right\|_{L_p(D_R)}.$$

Let $\delta := |\mathbf{x} - \mathbf{x}_0|$ and choose $R = \delta^\theta$ with $\theta \in (0, 1/2)$. Optimizing as usual gives

$$|u(\mathbf{x}) - u(\mathbf{x}_0)| \leq C\delta^\beta \left(\|u\|_{L_\infty(\Omega)} + \left\| \frac{f}{\rho_2^*(\mathbf{A})} \right\|_{L_p(\Omega)} \right),$$

for some $\beta > 0$. Thus u is Hölder continuous at every boundary point. Combining this boundary estimate with the interior Hölder regularity yields

$$u \in C^{0,\gamma}(\overline{\Omega}), \quad \gamma := \min\{\alpha, \beta\}.$$

This proves the claim. $\square$

B.2. Parabolic setting. Next, we turn to the parabolic setting. Let Ω , as before, satisfy the uniform exterior sphere condition, and define

$$Q_T := \Omega \times (0, T).$$

We decompose the parabolic boundary into the lateral, initial, and corner parts:

$$S_T := \partial\Omega \times (0, T), \quad B_0 := \Omega \times \{0\}, \quad S_0 := \partial\Omega \times \{0\}.$$

Let

$$u \in W_{\bar{p}}^{2,1}(Q_T) \cap C(\overline{Q_T}),$$

where we recall that the exponent $\bar{p}$ satisfies

$$\frac{d+2}{2} < \bar{p} < \infty.$$

Assume that u satisfies

$$(B.13) \quad \mathcal{P}u := \partial_t u - \sum_{i=1}^d a_i \partial_{x_i x_i} u - cu \leq f \quad \text{in } Q_T,$$

together with the boundary and initial conditions

$$(B.14) \quad u \leq 0 \quad \text{on } \partial_p Q_T := S_T \cup B_0 \cup S_0.$$

The space–time dependent diffusion matrix $\mathbf{A} := \text{diag}(a_1, \dots, a_d)$ satisfies the same ellipticity and Cordes conditions as in the elliptic case, namely (B.2) and (B.3), interpreted in the time-dependent setting. Moreover, we use the same notation as in (B.4) for the corresponding space–time dependent quantity.

Lemma B.6 (ABP estimate [5, Theorem 1]). *Let $u \in W_{\bar{p}}^{2,1}(Q_T) \cap C(\overline{Q_T})$ satisfy (B.13) and (B.14). Then we have the estimate*

$$(B.15) \quad \|u\|_{L_\infty(\Omega)} \leq C_{B.15} \left\| \frac{f}{(\rho_2^*(\mathbf{A}))^{(\bar{p}-1)/\bar{p}}} \right\|_{L_{\bar{p}}(Q_T)},$$

for some $C_{B.15}(Q_T, \bar{p}) > 0$.

Lemma B.7 (ABP estimate [5]). *Let $u \in W_{\bar{p}}^{2,1}(Q_T) \cap C(\overline{Q_T})$ satisfy (B.13) and (B.14). Then we have the estimate*

$$(B.16) \quad \|u\|_{L_\infty(\Omega)} \leq C_{B.16} |Q_T|^{2/(d+2)-1/\bar{p}} \left\| \frac{f}{(\rho_2^*(\mathbf{A}))^{(\bar{p}-1)/\bar{p}}} \right\|_{L_{\bar{p}}(Q_T)},$$

for some $C_{B.16}(d, \bar{p}, \sigma_1, \sigma_2, \lambda) > 0$.

Lemma B.8 (interior Hölder continuity for parabolic equations [5]). *Let*

$$Q_R(\mathbf{y}, s) := B_R(\mathbf{y}) \times (s - R^2, s] \subset Q_T,$$

and let

$$u \in W_{\bar{p}}^{2,1}(Q_R(\mathbf{y}, s)),$$

satisfy $\mathcal{P}u = f$ in $Q_R(\mathbf{y}, s)$. Then, for every concentric subcylinder

$$Q_{\sigma R}(\mathbf{y}, s) = B_{\sigma R}(\mathbf{y}) \times (s - \sigma^2 R^2, s], \quad 0 < \sigma < 1,$$

the following oscillation estimate holds:

$$(B.17) \quad \text{osc}_{Q_{\sigma R}(\mathbf{y}, s)} u \leq C \sigma^\alpha \left(\text{osc}_{Q_R(\mathbf{y}, s)} u + R^{2-(d+2)/\bar{p}} \left\| \frac{f}{\rho_2^*(\mathbf{A})} \right\|_{L_{\bar{p}}(Q_R(\mathbf{y}, s))} \right),$$

for some $\alpha(d, \sigma_1, \sigma_2, \lambda), C(R, d, \bar{p}, \sigma_1, \sigma_2, \lambda, \|c\|_{L_\infty}) > 0$. Consequently,

$$u \in C^{0,\alpha}(Q_{\sigma R}(\mathbf{y}, s)),$$

and

$$(B.18) \quad [u]_{C^{0,\alpha}(Q_{\sigma R}(\mathbf{y}, s))} \leq C R^{-\alpha} \left(\text{osc}_{Q_R(\mathbf{y}, s)} u + R^{2-(d+2)/\bar{p}} \left\| \frac{f}{\rho_2^*(\mathbf{A})} \right\|_{L_{\bar{p}}(Q_R(\mathbf{y}, s))} \right),$$

where the parabolic Hölder seminorm is defined by

$$[u]_{C^{0,\alpha}(Q)} := \sup_{\substack{(\mathbf{x}, t), (\bar{\mathbf{x}}, \bar{t}) \in Q \\ (\mathbf{x}, t) \neq (\bar{\mathbf{x}}, \bar{t})}} \frac{|u(\mathbf{x}, t) - u(\bar{\mathbf{x}}, \bar{t})|}{(|\mathbf{x} - \bar{\mathbf{x}}| + \sqrt{|t - \bar{t}|})^\alpha}.$$

In the parabolic setting, we again construct suitable barrier functions near the lateral, initial, and corner parts of the boundary. Combining these boundary and initial barriers with the interior Hölder estimate, we obtain the global Hölder continuity $u \in C^{0,\gamma}(\overline{Q_T})$, as stated in the following lemma.

Lemma B.9 (global Hölder continuity). *Let $u \in W_{\bar{p}}^{2,1}(Q_T) \cap C(\overline{Q_T})$ satisfy*

$$\mathcal{P}u = f \quad \text{in } Q_T, \quad u = 0 \quad \text{on } S_T \cup B_0 \cup S_0.$$

Then there exist $\gamma(d, \bar{p}, \sigma_1, \sigma_2, \lambda), C_{B.19}(Q_T, \bar{p}, \sigma_1, \sigma_2, \lambda, \|c\|_{L_\infty}) > 0$ such that

$$(B.19) \quad \|u\|_{C^{0,\gamma}(\overline{Q_T})} \leq C \left(\|u\|_{L_\infty(Q_T)} + \left\| \frac{f}{(\rho_2^*(\mathbf{A}))^{(\bar{p}-1)/\bar{p}}} \right\|_{L_{\bar{p}}(Q_T)} \right).$$

Proof. The interior Hölder continuity of u is already available from Lemma B.8, which ensures that

$$u \in C_{\text{loc}}^{0,\alpha}(Q_T,)$$

for some $\alpha > 0$. Therefore, it remains to establish Hölder continuity up to the boundary. We treat separately the three components of

$$\partial_p Q_T = S_T \cup B_0 \cup S_0.$$

Throughout the proof, the auxiliary parameters

$$\mu, \tau, R_1, R,$$

and the associated constants may vary from one step to another. For simplicity, we keep the same notation, although these quantities are chosen locally according to the barrier construction used in each part of the argument.

Step 1: Hölder decay near the lateral boundary S_T .

Fix a point

$$(\mathbf{x}_0, t_0) \in \partial\Omega \times (0, T].$$

Since $\partial\Omega$ satisfies the exterior sphere condition, there exist $R_0 > 0$ and

$$B_{R_0}(\mathbf{z}) \subset \mathbb{R}^d \setminus \Omega$$

such that

$$\overline{B_{R_0}(\mathbf{z})} \cap \overline{\Omega} = \{\mathbf{x}_0\}.$$

For $\mu > 0$ large, define

$$r(\mathbf{x}) = |\mathbf{x} - \mathbf{z}|, \quad \psi(\mathbf{x}) = R_0^{-\mu} - r(\mathbf{x})^{-\mu}.$$

A direct computation and choosing μ sufficiently large yield that there exist $R_1 > 0$ and $C_0 > 0$ such that

$$(B.20) \quad \mathcal{P}\psi \geq C_0 \quad \text{in } B_{R_1}(\mathbf{x}_0) \cap \Omega.$$

Now define

$$\Psi(\mathbf{x}, t) = \psi(\mathbf{x}) + \tau|\mathbf{x} - \mathbf{x}_0|^2 + \tau(t_0 - t),$$

with $\tau > 0$ small enough so that

$$\mathcal{P}\Psi \geq \frac{C_0}{2}.$$

Fix $0 < R < R_1$ and define

$$Q_{1,R} = (\Omega \cap B_R(\mathbf{x}_0)) \times (t_0 - R^2, t_0).$$

Define

$$v := u - M_R \Psi, \quad M_R := \frac{\|u\|_{L^\infty(Q_T)}}{\tau R^2}.$$

By construction,

$$\mathcal{P}v \leq |f| \quad \text{in } Q_{1,R}, \quad v \leq 0 \quad \text{on } \partial_p Q_{1,R}.$$

Hence the parabolic ABP estimate gives

$$\sup_{Q_{1,R}} |v| \leq CR^{2-(d+2)/\bar{p}} \left\| \frac{f}{(\rho_2^*(\mathbf{A}))^{(\bar{p}-1)/\bar{p}}} \right\|_{L_{\bar{p}}(Q_{1,R})}.$$

Therefore,

$$(B.21) \quad |u(\mathbf{x}, t)| \leq M_R \Psi(\mathbf{x}, t) + CR^{2-(d+2)/\bar{p}} \left\| \frac{f}{(\rho_2^*(\mathbf{A}))^{(\bar{p}-1)/\bar{p}}} \right\|_{L_{\bar{p}}(Q_{1,R})}.$$

Substituting the barrier estimate

$$\Psi(\mathbf{x}, t) \leq C(|\mathbf{x} - \mathbf{x}_0| + |t - t_0|^{1/2}).$$

this into (B.21) yields

$$|u(\mathbf{x}, t)| \leq C\|u\|_{L_{\infty}(Q_T)} \frac{|\mathbf{x} - \mathbf{x}_0| + |t - t_0|^{1/2}}{R^2} + CR^{2-(d+2)/\bar{p}} \left\| \frac{f}{(\rho_2^*(\mathbf{A}))^{(\bar{p}-1)/\bar{p}}} \right\|_{L_{\bar{p}}(Q_{1,R})}.$$

Let

$$\delta = |\mathbf{x} - \mathbf{x}_0| + |t - t_0|^{1/2}, \quad R = \delta^{\theta},$$

with $0 < \theta < 1/2$. Hence there exist $\beta_1 > 0$ and $C_1 > 0$ such that

$$(B.22) \quad |u(\mathbf{x}, t) - u(\mathbf{x}_0, t_0)| \leq C_1 \left(\|u\|_{L_{\infty}(Q_T)} + \left\| \frac{f}{(\rho_2^*(\mathbf{A}))^{(\bar{p}-1)/\bar{p}}} \right\|_{L_{\bar{p}}(Q_T)} \right) \delta^{\beta_1}.$$

Step 2: Hölder decay near the initial boundary B_0 .

Fix

$$(\mathbf{x}_0, 0), \quad \mathbf{x}_0 \in \Omega.$$

Choose $R > 0$ such that

$$B_R(\mathbf{x}_0) \subset \Omega,$$

and define

$$Q_{2,R} = B_R(\mathbf{x}_0) \times (0, R^2).$$

Consider the barrier

$$\Psi_2(\mathbf{x}, t) = t + \tau|\mathbf{x} - \mathbf{x}_0|^2.$$

For sufficiently small $\tau > 0$,

$$\mathcal{P}\Psi_2 \geq C_2 > 0.$$

Define

$$v = u - M_R \Psi_2, \quad M_R = \frac{\|u\|_{L_{\infty}(Q_T)}}{\tau R^2}.$$

Arguing exactly as before, we have

$$\mathcal{P}v \leq |f| \quad \text{in } Q_{2,R}, \quad v \leq 0 \quad \text{on } \partial_p Q_{2,R}.$$

Applying the parabolic ABP estimate and repeating the previous optimization argument yields

$$(B.23) \quad |u(\mathbf{x}, t) - u(\mathbf{x}_0, 0)| \leq C_3 \left(\|u\|_{L_\infty(Q_T)} + \left\| \frac{f}{(\rho_2^*(\mathbf{A}))^{(\bar{p}-1)/\bar{p}}} \right\|_{L_{\bar{p}}(Q_T)} \right) (|\mathbf{x} - \mathbf{x}_0| + t^{1/2})^{\beta_2}.$$

Step 3: Hölder decay near the corner boundary S_0 .

Fix

$$(\mathbf{x}_0, 0), \quad \mathbf{x}_0 \in \partial\Omega.$$

Using again the exterior sphere condition, define

$$\psi_3(\mathbf{x}) = R_0^{-\mu} - |\mathbf{x} - \mathbf{z}|^{-\mu},$$

and consider the barrier

$$\Psi_3(\mathbf{x}, t) = \psi_3(\mathbf{x}) + \tau|\mathbf{x} - \mathbf{x}_0|^2 + t.$$

As before, for suitable choices of μ and τ ,

$$\mathcal{P}\Psi_3 \geq C_4 > 0.$$

Setting

$$v := u - M_R \Psi_3, \quad M_R := \frac{\|u\|_{L_\infty(Q_T)}}{\tau R^2},$$

and applying the parabolic ABP estimate in

$$Q_{3,R} = (\Omega \cap B_R(\mathbf{x}_0)) \times (0, R^2),$$

we obtain

$$|u(\mathbf{x}, t)| \leq C \|u\|_{L_\infty(Q_T)} \frac{|\mathbf{x} - \mathbf{x}_0| + t^{1/2}}{R^2} + C R^{2-(d+2)/\bar{p}} \left\| \frac{f}{(\rho_2^*(\mathbf{A}))^{(\bar{p}-1)/\bar{p}}} \right\|_{L_{\bar{p}}(Q_{3,R})}.$$

Optimizing again with

$$R = \delta^\theta, \quad \delta = |\mathbf{x} - \mathbf{x}_0| + t^{1/2},$$

gives

$$(B.24) \quad |u(\mathbf{x}, t) - u(\mathbf{x}_0, 0)| \leq C_5 \left(\|u\|_{L_\infty(Q_T)} + \left\| \frac{f}{(\rho_2^*(\mathbf{A}))^{(\bar{p}-1)/\bar{p}}} \right\|_{L_{\bar{p}}(Q_T)} \right) \delta^{\beta_3}.$$

Finally, combining the interior Hölder continuity with the boundary estimates (B.22), (B.23), and (B.24), we conclude that

$$u \in C^{0,\gamma}(\overline{Q_T}),$$

for some

$$\gamma = \min\{\alpha, \beta_1, \beta_2, \beta_3\} > 0.$$

Moreover,

$$\|u\|_{C^{0,\gamma}(\overline{Q_T})} \leq C \left(\|u\|_{L_\infty(Q_T)} + \left\| \frac{f}{(\rho_2^*(\mathbf{A}))^{(\bar{p}-1)/\bar{p}}} \right\|_{L_{\bar{p}}(Q_T)} \right).$$

This completes the proof. $\square$

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